ICU Sepsis Risk Monitoring & Decision Support System

Ethan Vo
etvo@ucsd.edu

Guoxuan Xu
g7xu@ucsd.edu

Varun Pabreja
vpabreja@ucsd.edu

Yash Patel
ypatel@ucsd.edu

Kyle Shannon
kmshannon@ucsd.edu

Abstract

This report aims to investigate the necessity, utility, and consequences of an in-ICU Sepsis Risk Monitoring System. It does so by rigorously exploring key bottlenecks in sepsis-related clinician workflows, and proposing a production-level system to solve some (if not all) of them. The paper further argues the integration of a state-of-the-art supervised machine learning model into the prediction process based on measured success in some critical areas related to sepsis decision-making. Finally, key outcomes and measures of improvement have been recorded.

1 Introduction

1.1 Problem Scope

Sepsis is a life-threatening condition characterized by a dysregulated immune response to infection, typically leading to organ dysfunction and, if untreated, death. Moreover, it is quite costly to treat (both in terms of time and money). In the United States alone, sepsis contributes to over 270,000 deaths, with sepsis-related hospital costs exceeding $38 billion per year (Hollenbeak et al. 2023). Within the intensive care unit (ICU), where the most critically ill patients receive care, sepsis represents both a common admission diagnosis and a frequent complication of hospitalization.

Early recognition and timely intervention are paramount in sepsis care. The Surviving Sepsis Campaign guidelines emphasize the importance of identifying sepsis within the first hour and initiating treatment bundles as quickly as possible (Evans et al. 2021). Each hour of delay in appropriate antibiotic administration is associated with increased mortality risk. However, despite the clinical urgency, sepsis detection remains challenging in practice due to numerous limitations (as inspired by anecdotal evidence from interviews conducted with authors of related works):

1. **Delayed Alerts**: Traditional scoring systems (e.g., SIRS, qSOFA) often trigger only after significant physiological decompensation has occurred, reducing the window for intervention, increasing the likelihood of patient deterioration.
2. **Limited Interpretability**: Machine learning models trained on ICU data may achieve high predictive performance but often function as "black boxes", making it difficult for clinicians to trust or act on their predictions without understanding the reasoning.
3. **Poor Information Flow**: ICU clinicians face information overload from different monitoring systems. Relevant data for sepsis detection - vitals trends, laboratory results, medication orders - are often siloed across different systems, requiring clinicians to mentally integrate disparate information sources for time-critical decision-making.
4. **Workflow Misalignment**: Existing decision support tools are rarely designed with clinical workflows in mind, failing to account for the distinct information needs of bedside nurses and attending physicians.
5. **Alert Fatigue**: Clinicians often experience alert fatigue due to consistent monitoring of different patient information, with most of this information not being critical to sepsis decision-making.

These challenges motivate the need for a data product that not only predicts sepsis risk accurately but also integrates seamlessly into ICU workflows, presents information transparently, and supports different clinical roles.

1.2 Motivating Question

Our project addresses the following motivating question:

> *Can we design a data product to improve information flow and provide just-in-time predictions for sepsis patients that could provide benefits to patients and hospital ICU medical staff?*

This question encompasses several key dimensions:

- **Design**: What system architecture, user interface, and data presentation best support clinical decision-making around sepsis?
- **Information flow**: How can we integrate and synthesize data from multiple sources (vitals, labs, medications, procedures) to provide actionable insights without adding to information overload?
- **Just-in-time predictions**: What prediction horizon (e.g., 6 hours ahead) and update frequency (e.g., hourly) balance clinical utility with computational feasibility and alert fatigue?

1.3 Connection to Prior Work

This project builds directly on our team's previous work with the MIMIC-III database. In Quarter 1, we successfully replicated a study on multimorbidity phenotyping in ICU patients, demonstrating our ability to work with complex clinical data, engineer features, train machine learning models, and evaluate results in the context of critical care. The previous project focused on understanding patient heterogeneity through chronic comorbidity patterns.

The current project extends this foundation in 2 crucial ways:

1. **Real-Time Predictions**: While our multimorbidity work was retrospective, this in-ICU sepsis project requires modeling time-evolving risk and considering prediction horizons relevant to clinical decision-making.
2. **Productionizing Research Design**: Rather than replicating published findings, we are now designing a novel data product with explicit consideration of clinician needs.

By leveraging our existing MIMIC-III expertise while addressing these new challenges, we aim to create a compelling demonstration of how data science can translate research insights into practical clinical tools.

2 Methods

2.1 Data Source and Cohort Definition

Our data comes from the MIMIC-IV Clinical Database - a large, publicly available dataset of de-identified health records collected from ICU admissions at Beth Israel Deaconess Medical Center. The database contains a wide range of clinical information including patient demographics, vital signs, laboratory results, medication records, procedure events, diagnosis codes, and clinical notes. We chose to use MIMIC-IV for this project because the volume of data it provides is large enough to support training and evaluating machine learning models, and its reliability is underscored by its presence as a dataset in sepsis research.

From this database, we selected a study cohort based on the following criteria:

1. **adult patients (age $\geq$ 18)**,
2. **first ICU admission only**,
3. **ICU length of stay greater than 12 hours**,
4. **no sepsis diagnosis present at the time of admission**,
5. **no missing vital sign records**

The main reason we restricted the cohort to first admissions only is because we do not want to introduce statistical dependence between observations. Patients may have multiple stays in the ICU for the same hospital admission, and information about prior diagnostics can and do affect current diagnostics. We also require that each patient had at least 12 hours of ICU data recorded so that the trained model can utilize clinical observations to produce meaningful output. If a patient stayed only for a few hours, there would not be enough

data to generate a meaningful prediction (at least in the case of sepsis). In addition, we excluded patients who already had a sepsis diagnosis at the time of admission. The goal of our system is to predict new-onset sepsis, meaning sepsis that develops during the ICU stay, rather than to detect sepsis that is already present when the patient first enters the ICU. After applying all of these filters, we expect the final cohort to contain approximately 20,000 to 30,000 ICU admissions.

2.2 Label Definition

We defined sepsis onset using the Sepsis-3 criteria. Under Sepsis-3, a patient is considered to have sepsis when two conditions are met - organ dysfunction and suspected infection. Suspected infection is identified by the administration of new antibiotics together with the collection of body fluid cultures (blood, urine, respiratory specimens, etc.) within a ±24 hour window. In other words, if a clinician orders antibiotics and also collects a culture sample around the same time, it counts as a suspected infection event. Organ dysfunction is defined as an acute increase in the Sequential Organ Failure Assessment (SOFA) score of ≥2 points from the patient's baseline SOFA score.

The onset timestamp is set to whichever event came first, either the antibiotic order or the culture collection, as long as both the infection and organ dysfunction criteria were satisfied within the time window. We computed baseline SOFA scores from data collected in the first 24 hours of the ICU stay. If pre-ICU values were available, we used those instead.

2.3 System Architecture

The system follows a three-layer architecture. The full architecture is documented in the project repository (`https://github.com/kshannon-ucsd/icu-sepsis-decision-support`), however, an image is provided to make following the description more intuitive.

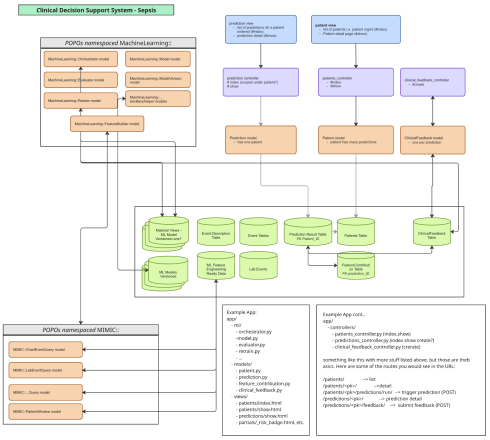

Figure 1: MVC Flowchart of the in-ICU data product

The bottom layer is the data ingestion pipeline. This pipeline pulls raw patient data from MIMIC-IV and loads it into the relational database tables described in Section 2.4. It is responsible for extracting the raw records, cleaning them, and aggregating the clinical events into hourly intervals. The hourly aggregation is important because the rest of the system operates on an hourly time step, so all of the raw data needs to be bucketed into one-hour windows before it can be used by the model or displayed to the user.

The middle layer is the machine learning pipeline. This pipeline reads from the database tables, applies feature transformations, and then trains and runs the predictive models. Once the models produce sepsis risk scores, those scores are written back to the database so that the user can access them.

The top layer is the frontend. It consists of three clinical views, which are described in more detail in Section 2.5. These views are served through Django controllers and URL routes that query the database for patient data and prediction results. Each view is built for a different clinical role. Because different roles need different levels of detail, the information shown in each view is different.

2.4 Database Schema

We designed a six-table relational schema to store patient data and to support the prediction pipeline. The schema is organized around one central patient profile table, with five hourly event tables linked to it. Each of the hourly tables stores a different category of clinical measurements recorded on an hourly basis. These hourly tables also serve as the engineered feature set that the machine learning models read from. Each row in a given hourly table represents one patient at one hour, and it contains the clinical measurements that were recorded for that patient during that hour. A description of the tables has been provided below the image.

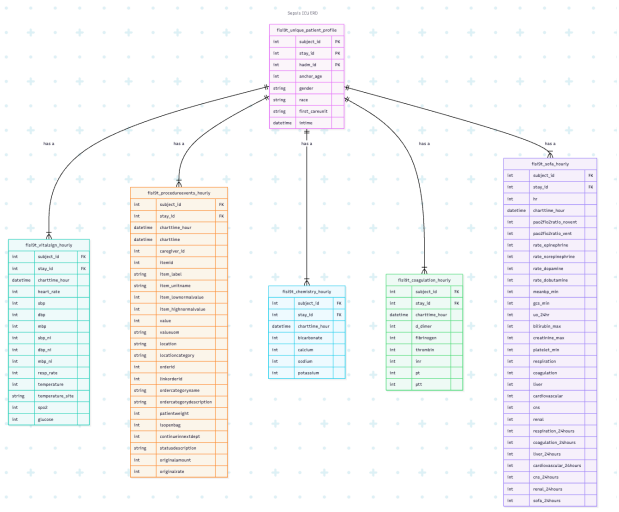

Figure 2: ERD of the database schema

The tables are as follows:

1. `fsi9t_unique_patient_profile`: This is the parent table. It stores patient demographics and admission information. The columns include `subject_id`, `stay_id`, `hadm_id`, age, gender, race, first care unit, and admission time. Every patient in the cohort has exactly one row in this table.
2. `fsi9t_vitalsign_hourly`: This table stores hourly vital sign measurements. The vital signs recorded include heart rate, systolic blood pressure, diastolic blood pressure, mean blood pressure (both invasive and non-invasive), respiratory rate, temperature, SpO2, and glucose.
3. `fsi9t_procedureevents_hourly`: This table stores hourly procedure event records. Each row represents a procedure that occurred during a given hour, and includes the item identifier, the item label, the measured value, the unit of measurement, order category information, and caregiver information.
4. `fsi9t_chemistry_hourly`: This table stores hourly chemistry lab results. The lab values recorded are bicarbonate, calcium, sodium, and potassium.
5. `fsi9t_coagulation_hourly`: This table stores hourly coagulation lab results. The lab values recorded are D-dimer, fibrinogen, thrombin, INR, PT, and PTT.
6. `fsi9t_sofa_hourly`: This table stores hourly SOFA scores. The scores are broken down by organ system, specifically respiratory, coagulation, liver, cardiovascular, CNS, and renal. The table also stores the underlying clinical values that are used to compute each component score, such as vasopressor rates, GCS, urine output, bilirubin, creatinine, and platelet count. In addition, the table includes 24-hour rolling SOFA sub-scores for each organ system and the overall 24-hour SOFA score.

All five hourly tables reference the patient profile table through `subject_id` and `stay_id` foreign keys. This creates a one-to-many relationship, meaning that one patient can have many hourly records across each of the five tables. Each hourly table also includes a `charttime_hour` column that marks the hour of the observation. This column is what we use to align data across tables for a given patient at a given time. For example, if we want to look at all the data for a specific patient at hour 5 of their stay, we can join across the tables using `subject_id`, `stay_id`, and `charttime_hour`.

2.5 Frontend Design

The frontend is built as a Django web application with three fully implemented clinical views. What follows is a description of the intended users and the general purpose of each view.

The first page is a patient list page, and it serves as the management view. This view is intended for ICU charge nurses, unit managers, and ICU team leaders. The page displays a paginated table that shows all patients currently in the ICU. Each row in the table shows the patient's **subject ID, stay ID, age, gender, race, care unit, and admission time**. The page also includes a simulation clock that advances time in one-hour increments. Every time the clock is advanced, patients who were admitted during that hour appear in the table. The goal of this view is to give staff a unit-wide picture of who is currently in the ICU so they can

see all their patients at a glance and prioritize attention and allocate resources accordingly.

The second page is a patient detail page, and it serves as the patient view. This view is intended for bedside nurses, attending physicians, and ICU fellows. When a clinician selects a patient from the patient list, they are taken to this page, which shows all of that patient's clinical data in one place. A patient's demographic information, vital signs chart (i.e. the patient's heart rate, blood pressure, SpO2, respiratory rate, temperature, glucose etc. plotted using Plotly), log of procedures performed on the patient and when they occurred, and a prediction panel are shown. This panel queries the machine learning pipeline through a JSON API endpoint and displays the patient's current sepsis risk score along with their comorbidity group. The purpose of this view is to support bedside decision-making by giving clinicians a single page where they can see everything they need to know about a patient's current clinical status and sepsis risk.

The third page is a prediction history view intended for attending physicians, researchers, and quality improvement teams. This view shows how a patient's sepsis risk score changed over their entire ICU stay, so clinicians and researchers can review the full trajectory at any time to make in depth analysis. Aside from spesis probability score, the view surfaces a set of supporting clinical values to provide context for each hourly prediction, including ventilation-adjusted oxygenation ratios (PaO2/FiO2 for both non-ventilated and ventilated states) and vasopressor infusion rates (epinephrine, norepinephrine, dopamine, and dobutamine). The view also displays a panel of discrete organ function values including minimum mean arterial pressure, minimum GCS, maximum bilirubin, maximum creatinine, and minimum platelet count, alongside 24-hour rolling SOFA sub-scores for each organ system and a total SOFA score. These metrics explain the risk score to clinicians. By surfacing the underlying organ function indicators, the view helps clinicians understand the clinical drivers behind each prediction. For example, a rising creatinine paired with a high renal SOFA sub-score signals kidney deterioration, giving the clinician actionable context rather than just a number. Lastly, the view presents a set of historically similar cases retrieved from the database using cosine similarity over the patient's feature vector, giving clinicians a reference point for how comparable patients progressed.

3 Results

3.1 Model Performance

The sepsis prediction model is a Random Forest classifier trained on 2,079,753 hourly observations derived from the MIMIC-IV cohort, using 53 features (49 numeric, 4 categorical) including the patient's latent comorbidity class. Patients were partitioned by `stay_id` using an 80/20 grouped shuffle split, yielding 1,661,817 training observations and 417,936 test observations.

It is important to note that the positive label rate in this dataset is 71.3%. This does not mean that 71% of patients develop sepsis. Rather, once a patient meets the Sepsis-3 onset criteria at a given hour, all subsequent hours in that stay are assigned a positive label.

This forward-propagation reflects the clinical reality that sepsis is an ongoing state once it develops, and it is the convention used by the prediction pipeline to assess whether a patient is currently at risk.

The model achieved a test ROC-AUC of **0.848** and a PR-AUC of 0.927. The ROC-AUC measures the model's ability to correctly rank a septic hour above a non-septic one across all possible thresholds; a value of 0.848 compares favorably to traditional clinical scoring systems such as qSOFA, which typically achieves an AUROC of approximately 0.62 in comparable ICU settings. The PR-AUC of 0.927, while high, should be interpreted in light of the elevated base rate: a random classifier would achieve a PR-AUC of approximately 0.713, so the model's true gain over baseline is approximately 0.21 points.

Table 1 reports precision, recall, specificity, and F1 at four candidate decision thresholds.

Table 1: Model performance at four decision thresholds on the held-out test set.

Threshold	Precision	Recall	Specificity	F1
0.10	0.711	1.000	0.008	0.831
0.20	0.756	0.984	0.228	0.855
0.30	0.811	0.931	0.472	0.867
0.50	0.893	0.749	0.781	0.815

Precision: fraction of flagged hours that were truly septic; *Recall*: fraction of actual sepsis hours caught; *Specificity*: fraction of non-septic hours correctly left unflagged; *F1*: harmonic mean of precision and recall.

A threshold of 0.30 yields the highest F1 score (0.867) and is the recommended operating point, balancing precision (0.811) and recall (0.931). However, its specificity of 0.472 means that more than half of non-septic hours still trigger an alert, which remains the model's primary clinical weakness. For settings where missing a sepsis case is considered more dangerous than generating a false alert, a threshold of 0.20 may be preferred, as it achieves near-perfect recall (0.984) at the cost of further reduced specificity (0.228).

3.2 System Demonstration

The deployed system allows clinicians to navigate freely between all three views. The management view provides a unit-wide patient list, and selecting any patient transitions directly to that patient's detail view. From the detail view, clinicians can access the prediction history view to review the full risk trajectory for that stay.

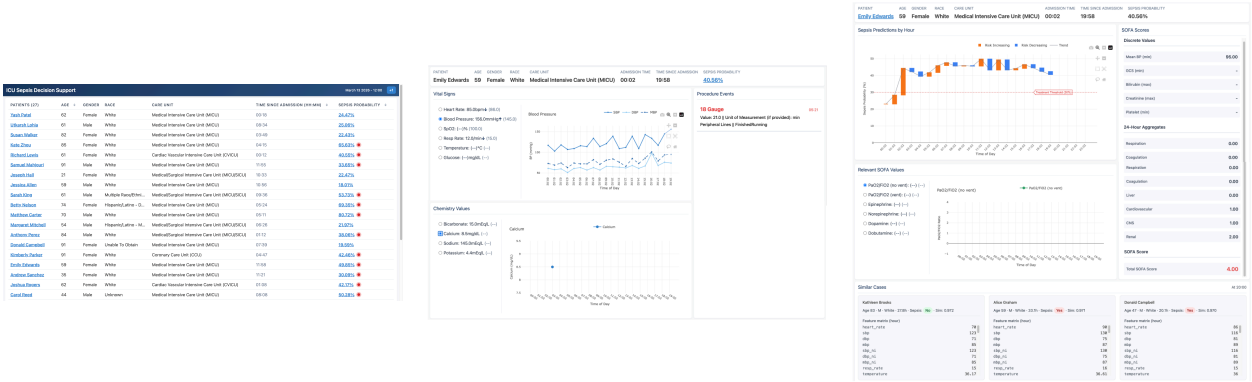

Figure 3: Screenshots of the three clinical views: Management View (left), Patient Detail View (center), and Prediction History View (right).

The prediction pipeline is decoupled from the frontend through a JSON API endpoint. This design means that the underlying model can be swapped or updated without requiring changes to the clinical views, providing flexibility to adopt improved models as they become available.

4 Discussion

4.1 Limitations

First, the model was trained and evaluated exclusively on MIMIC-IV, a single-center retrospective dataset from Beth Israel Deaconess Medical Center. Performance on patient populations from other hospital systems or geographic regions is unknown and may differ. Second, no prospective validation has been conducted; the model has not been tested in a live clinical environment, and its real-world utility remains unverified. Third, the current system is not connected to a live EHR feed, instead we simulate the processing via a data pipeline. Thus, deploying this system in a hospital would require substantial additional integration work. Finally, the Sepsis-3 label definition introduces some noise, as the criteria rely on antibiotic and culture ordering patterns that may vary across clinicians and institutions.

4.2 Ethical Considerations

Reducing a patient's clinical state to a single sepsis probability score risks oversimplifying a complex medical judgment, and clinicians may anchor on that number without adequately weighing other contextual factors. To mitigate this, the system surfaces an abundance of supporting metrics alongside each prediction — organ function indicators, SOFA subscores, and so on, so that the risk score serves as one input to clinical reasoning rather than a directive. The system is designed to augment, not replace, clinical decision-making.

5 Conclusion

This project delivers a production-level ICU sepsis decision support system designed to improve information flow and clinical decision-making around sepsis. The core contribution is a three-view Django frontend built around distinct clinical needs. The management view gives charge nurses and unit leaders a consolidated, real-time picture of all patients in the ICU, reducing the overhead of tracking patient status across disparate systems. The patient detail view gives bedside clinicians a single page with live sepsis risk scores, vital sign trends, procedure logs, and comorbidity context, directly addressing the information fragmentation problem. The prediction history view gives attending physicians hour-by-hour retrospective access to a patient's risk trajectory alongside organ function indicators and similar historical cases, supporting post-hoc clinical review and decision accountability. Underpinning all three views is a trained Random Forest sepsis prediction model that achieves a ROC-AUC of 0.848, meaningfully outperforming traditional scoring systems such as qSOFA.

5.1 Future Work

Several directions would strengthen the system in future iterations. On the frontend side, the clinical views would benefit from resizing components based on the significance of each displayed feature, so that these components receive visual priority. Additionally, the similar-patient retrieval currently relies on unweighted cosine similarity; replacing this with a more advanced similarity calculation such as weighted similarity would improve the relevance of historical case matches. On the data side, the pipeline is currently built around the MIMIC-IV dataset; connecting it to a live EHR feed would require replacing the batch ingestion layer with a real-time data stream and handling the latency and missingness patterns that accompany live clinical data. On the model side, imaging data such as MIMIC-CXR with radiographic sign of infection and organ dysfunction could potentially improve spesis prediction accuracy.

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